Emergency Evacuation Planner:

PEAS

Performance	1. The way to reach faster 3.adapt to changes	2. The shortest way
Environment	1. Streets 3. Roads	2. Office 4. Buildings
Actuators	1. Microphone 3.guidance lights that direct people to safe exits	2. Robots give people the ways
Sensors	1. Camera 3. GPS	2. Fire sensor 4.Smoke detectors

Environment (ODESDA):

O (Fully / Partially Observable)	Partially : The system cannot see or monitor everything at the same time
D (Deterministic / Stochastic)	Stochastic: The next state cannot be determined with certainty. The fire may spread unexpectedly, smoke levels may increase, or people may take unpredictable or incorrect actions.
E (Episodic / Sequential)	Sequential: Each decision affects the next one. For example, choosing one evacuation route may influence the available options and safety in the next steps.
S (Static / Dynamic)	Dynamic:Environment changes over time new dangers may appear.
D (Discrete / Continuous)	Continuous: There are no fixed or discrete states; positions and events can change smoothly as people move
A (Single / Multi-Agent)	Mulit: More than one agent (person or sensor system) acts at the same time, and their actions may affect each other

Problem Formulation:

State	A state represents the positions of all people in the building, the condition of each exit (open or blocked), and the status of hazards at a given time
Initial State	All employees are located inside the office building in their respective rooms or areas. Some exits may be open or closed, and some areas might be affected by fire
Successor Function	Defines all possible actions that can be taken from a given state for example: - A person moves from one room or corridor to another if it's not blocked. - A door is opened or unlocked. - An alarm or instruction is triggered to guide people. Each action leads to a new state of the environment
Goal Test	All employees have successfully reached a safe exit
Path Cost	The system aims to find a plan that evacuates all people to safety in the shortest possible time while minimizing exposure to danger and avoiding crowded areas

Team Members:

- Sara Mahmud Abdelrahman group:A (23013056)
- Nada Mohamed Zainelaabedin Afia group:c (23012744)
- Menna khaled Abdelrehem group:c (23012753)